

Ejercicios de la Parte 1: Capítulo 1.

2. Given an integer $N \in \mathbb{N}$, $N \geq 1$, and N^2 continuous functions, $a_{ij} \in \mathcal{C}(\mathbb{R}; \mathbb{R})$, $i, j \in \{1, \dots, N\}$, consider the Cauchy problem

$$\begin{cases} u'_i = \sum_{j=1}^N a_{ij}(t) \cos u_j, & i \in \{1, \dots, N\}, \\ u(t_0) = u_0. \end{cases}$$

Prove that, for every $t_0 \in \mathbb{R}$ and $u_0 \in \mathbb{R}^N$, it admits a unique solution, $u(t; t_0, u_0)$, globally defined in $t \in \mathbb{R}$.

3. Consider the following Cauchy problem:

$$\begin{cases} u' = a(t)u + b(t)\frac{u^n}{1+u}, \\ u(0) = u_0, \end{cases}$$

with $a, b \in \mathcal{C}([0, +\infty); \mathbb{R})$, $n \geq 1$ a natural number, and $u_0 \in (-1, 0)$. Analyze the possible behaviors of the maximal solution to the right.

4. Prove that, for every $t_0 \in \mathbb{R}$ and $(u_0, v_0) \in \mathbb{R}^2$, the Cauchy problem

$$\begin{cases} u' = t \cos t \sin v, \\ v' = t \sin t \cos u, \end{cases} \quad (u(t_0), v(t_0)) = (u_0, v_0),$$

has a unique solution globally defined in $t \in \mathbb{R}$.

5. Prove that, for every $t_0 > 0$ and $(u_0, v_0) \in \mathbb{R}^2$, the problem

$$\begin{cases} u' = \cos t \log t \sin^2 v, \\ v' = e^t \sin t \cos^2 u, \\ (u(t_0), v(t_0)) = (u_0, v_0), \end{cases}$$

has a unique solution globally defined in $t \in (0, +\infty)$.

6. Show that the problem

$$\begin{cases} u' = u^2 + v, \\ v' = v^2 + u, \\ (u(0), v(0)) = (1, 1), \end{cases}$$

has a unique maximal solution to the right and to the left. Prove that the former blows up after a finite time. Determine the maximal existence time of the solution. What is the global backward behavior of the solution?

[Hint: Observe that u and v play the same role in the system.]

7. Given $\lambda > 0$, $\mu > 0$, $b > 0$, and $c > 0$, consider the competing species model

$$\begin{cases} u' = \lambda u - u^2 - buv, \\ v' = \mu v - v^2 - cuv, \\ (u(0), v(0)) = (u_0, v_0), \end{cases} \quad (5.60)$$

where $u_0 \geq 0$ and $v_0 \geq 0$.

- Prove that (5.60) has a unique maximal solution to the right, (I, u, v) .
 - Determine the maximal solution when $u_0 = 0$ or $v_0 = 0$.
 - Prove that $u(t) > 0$ and $v(t) > 0$ for all $t \in I$ if $u_0 > 0$ and $v_0 > 0$.
 - Prove that $I = [0, \infty)$ if $u_0 > 0$ and $v_0 > 0$.
8. Repeat the previous exercise for the predator-prey model, which is given by

$$\begin{cases} u' = \lambda u - u^2 - buv, \\ v' = \mu v - v^2 + cuv, \\ (u(0), v(0)) = (u_0, v_0), \end{cases}$$

where $\lambda > 0$, $\mu > 0$, $b > 0$, $c > 0$, $u_0 \geq 0$, and $v_0 \geq 0$.

9. Given $\lambda > 0$ and $b > 0$, consider the symbiotic model

$$\begin{cases} u' = \lambda u - u^2 + buv, \\ v' = \lambda v - v^2 + buv, \\ (u(0), v(0)) = (1, 1). \end{cases}$$

Show that it admits a unique maximal solution to the right. Determine this solution and its maximal existence time in terms of the parameter $b > 0$, which measures the synergy between the individuals of the species u and v .

10. Given the planar system

$$\begin{cases} u' = u + v - u(u^2 + v^2), \\ v' = -u + v - v(u^2 + v^2), \end{cases}$$

obtain an equivalent one using polar coordinates,

$$u = \rho \cos \theta, \quad v = \rho \sin \theta.$$

Solve the resulting system and represent the solutions of the original one in the (u, v) -plane.

11. Given $p > 0$ and $q > 0$ with $pq < 1$, consider the problem

$$\begin{cases} u' = |v|^p, \\ v' = |u|^q, \\ (u(0), v(0)) = (1, 1). \end{cases}$$

Show the existence of a unique maximal solution to the right, (I, u, v) , and analyze its global character.

12. Given $h, j, k, \ell \in \mathbb{N} \setminus \{0\}$, prove that, for every $t_0 \in \mathbb{R}$ and $(u_0, v_0) \in \mathbb{R}^2$, the problem

$$\begin{cases} u' = -u^{2h+1}v^{2\ell}, \\ v' = -u^{2j}v^{2k+1}, \\ (u(t_0), v(t_0)) = (u_0, v_0), \end{cases}$$

has a unique solution, which is globally defined in $[t_0, \infty)$.

13. A function $f \in \mathcal{C}^1(\mathbb{R} \times \mathbb{R}^N; \mathbb{R}^N)$ is said to be of the *Kolmogorov type* if, by denoting

$$u = (u_1, \dots, u_N)^T \quad \text{and} \quad f = (f_1, \dots, f_N)^T,$$

there exist functions $g_i \in \mathcal{C}(\mathbb{R} \times \mathbb{R}^N; \mathbb{R})$, $1 \leq i \leq N$, such that

$$f_i(t, u) = g_i(t, u)u_i \quad \text{for all } i \in \{1, \dots, N\}, t \in \mathbb{R}, u \in \mathbb{R}^N.$$

Assuming that f is of the Kolmogorov type, consider, for every $u_0 \in \mathbb{R}^N$, the initial value problem

$$\begin{cases} u' = f(t, u), \\ u(0) = u_0. \end{cases} \quad (5.61)$$

- (a) Prove the existence of a unique maximal solution of (5.61) and discuss all its admissible behaviors.
(b) Let $(I, u(\cdot; u_0))$ be the maximal solution of (5.61). Prove that

$$\text{sign } u_i(t; u_0) = \text{sign } u_{0i}$$

for all $t \in I$ and $i \in \{1, \dots, N\}$, where $u_0 = (u_{01}, \dots, u_{0N})^T$.

- (c) Suppose that, in addition,

$$g_i(t, u) = \lambda_i - \sum_{j=1}^N a_{ij}u_j, \quad 1 \leq i \leq N,$$

where $\lambda_i \in \mathbb{R}$ and $a_{ij} \geq 0$ are constant for each $i, j \in \{1, \dots, N\}$. Show that, in such a case, $u(t; u_0)$ is globally defined on $[0, \infty)$ provided $u_{0i} \geq 0$ for all $i \in \{1, \dots, N\}$.

Assuming, moreover, that $\lambda_i < 0$ for every $i \in \{1, \dots, N\}$, show that, for any of those i 's,

$$\lim_{t \uparrow \infty} u_i(t) = 0.$$

- (d) Give examples, with $N = 2$ and $a_{12} = a_{21} < 0$, for which the solution blows up in a finite time.

14. Given an integer $N \geq 1$ and a non-zero square matrix

$$A = (a_{ij})_{1 \leq i, j \leq N} \in \mathcal{M}_N(\mathcal{C}(\mathbb{R}; \mathbb{R})),$$

consider the function

$$f(t, u) = (f_1(t, u), \dots, f_N(t, u))^T, \quad u = (u_1, \dots, u_N)^T,$$

whose components are defined as

$$f_i(t, u) = \sum_{j=1}^N a_{ij}(t) |u_j|^{p_{ij}}, \quad (t, u) \in \mathbb{R} \times \mathbb{R}^N, \quad 1 \leq i \leq N,$$

where, for every $i, j \in \{1, \dots, N\}$, $p_{ij} \geq 1$. Prove the existence and uniqueness of a maximal solution for the problem

$$\begin{cases} u' = f(t, u), \\ u(0) = u_0 \in \mathbb{R}^N. \end{cases}$$

Discuss all its admissible behaviors, and give examples for all of them in the special case of $N = 2$ by choosing appropriate $a_{ij}(t)$ and p_{ij} .

15. Prove that, for every $\varepsilon \geq 0$, the problem

$$\begin{cases} u'(t) = t [\sin u(t) + \varepsilon \cos u(t)], \\ u(0) = 1, \end{cases}$$

has a unique solution globally defined in $[0, \infty)$, denoted by u_ε , and estimate $|u_\varepsilon(t) - u_0(t)|$ for every $t > 0$ and $\varepsilon > 0$.

16. Prove that, for every $\varepsilon \geq 0$, the problem

$$\begin{cases} u'(t) = t \sin u(t), \\ u(0) = \varepsilon, \end{cases}$$

has a unique solution globally defined in $[0, \infty)$, denoted by u_ε , and estimate $|u_\varepsilon(t) - u_0(t)|$ for every $t > 0$ and $\varepsilon > 0$.